

In this paper, every Ring is Commutative Ring with $1 \neq 0$.

Thm. Let $A_1, \dots, A_k$ be Ideals in R . Define a map

$$\mathcal{Q}: R \rightarrow R/A_1 \times \dots \times R/A_k: r \mapsto (r+A_1, \dots, r+A_k)$$

is a Ring Homomorphism with $\ker \mathcal{Q} = A_1 \cap A_2 \cap \dots \cap A_k$.

If for each $1 \leq i < j \leq k$, $A_i + A_j = R$ (comaximal), then $\mathcal{Q}$ is onto and

$$\ker \mathcal{Q} = A_1 \cap \dots \cap A_k = A_1 A_2 \dots A_k.$$

Therefore, $R/(A_1 \dots A_k) = R/(A_1 \cap \dots \cap A_k) \cong R/A_1 \times \dots \times R/A_k$.

Proof. First, prove this for $k=2$. The map

$$\mathcal{Q}: R \rightarrow R/A_1 \times R/A_2: r \mapsto (r+A_1, r+A_2)$$

is just the natural projection for $R/A_1, R/A_2$ elementwise, so ring-homomorphism.

Moreover,

$$\begin{aligned} \mathcal{Q}(r) = 0 &\Leftrightarrow r+A_1 = A_1 \text{ and } r+A_2 = A_2 \\ &\Leftrightarrow r \in A_1 \text{ and } r \in A_2 \end{aligned}$$

so $\ker \mathcal{Q} = A_1 \cap A_2$. Now, suppose that $A_1 + A_2 = R$. Since $1 \in R = A_1 + A_2$,
 $x+y=1$ for some $x \in A_1$ and $y \in A_2$.

Therefore, $\mathcal{Q}(x) = \mathcal{Q}(1-y) = (x+A_1, 1-y+A_2) = (0+A_1, 1+A_2)$.

And similarly $\mathcal{Q}(y) = (1, 0)$. Now, given $(r_1+A_1, r_2+A_2) \in R/A_1 \times R/A_2$,

$$\begin{aligned} \mathcal{Q}(r_2x + r_1y) &= \mathcal{Q}(r_2)\mathcal{Q}(x) + \mathcal{Q}(r_1)\mathcal{Q}(y) \\ &= \mathcal{Q}(r_2)(\bar{0}, 1) + \mathcal{Q}(r_1)(1, \bar{0}) \\ &= (\bar{0}, \bar{r}_2) + (\bar{r}_1, \bar{0}) = (r_1+A_1, r_2+A_2). \end{aligned}$$

Thus $\mathcal{Q}$ is onto. And $A_1 A_2 \subseteq A_1 \cap A_2$, because: for $a_i \in A_1, b_i \in A_2$,

$$a_1 b_1 + \dots + a_n b_n \in A_1 \text{ and } A_2 \text{ since } A_1, A_2 \text{ are Ideal.}$$

Meanwhile, if $A_1 + A_2 = R$, for any $c \in A_1 + A_2$,

$$c = c \cdot 1 = c(x+y) = cx + cy \in A_1 + A_2.$$

So proved.

Thm. Let R be a Ring with identity $1 \neq 0$.

Let $e \in R$ be an idempotent in R such that $er = re$ for all $r \in R$.

Then, Re and $R(1-e)$ are (two-sided) ideal of R and that

$$R \cong Re \times R(1-e)$$

Moreover, e and $1-e$ are identity for Re and $R(1-e)$, respectively.

Proof. Notice e commutes with respect to all elements in R , $Re = eR$.

and given $r_1, r_2 \in R$, $r_1e - r_2e = (r_1 - r_2)e \in Re$, so Re is subgroup

and given $r \in R$ and $e \in Re$, $re \in Re$, so Re is left-sided ideal.

Since $eR = Re$, similarly it is right-sided ideal.

Moreover, $1-e$ also commutes in R , so $R(1-e)$ is ideal, similarly Re .

Given $r \in R$, $r = re + r - re$, so $R = Re + R(1-e)$ maximal.

By the Chinese Remainder Theorem, $Re \cap R(1-e) = Re(1-e)$.

observe that: given $r_1, r_2 \in R$,

$$r_1e \cdot (r_2 - r_2e) = r_1r_2e - r_1r_2e^2 = r_1r_2(e - e^2) = r_1r_2 \cdot 0 = 0.$$

Therefore, $Re \cdot R(1-e) = \{0\}$. Meanwhile,

$$\mathcal{Q}: R \rightarrow Re; r \mapsto re$$

is onto Ring-homomorphism, because: Surjective is trivial and

$$\mathcal{Q}(r_1 + r_2) = (r_1 + r_2)e = r_1e + r_2e = \mathcal{Q}(r_1) + \mathcal{Q}(r_2) \text{ and}$$

$$\mathcal{Q}(r_1r_2) = r_1r_2e = r_1r_2e^2 = r_1e r_2e = \mathcal{Q}(r_1)\mathcal{Q}(r_2)$$

And $\mathcal{Q}(r) = re = 0 \Leftrightarrow r \in R(1-e)$, because

$$re = 0 \Rightarrow r = (r-e) - (r-e) \cdot e \in R(1-e)$$

$r \in R(1-e) \Rightarrow r = r' - r'e$ for some $r' \in R \Rightarrow re = r'e - r'e^2 = 0$. So $\ker \mathcal{Q} = R(1-e)$.

So, $Re \cong R/R(1-e)$ and similarly $R(1-e) \cong R/Re$.

Consequently, the Chinese Remainder theorem gives

$$R \cong R/\{0\} \cong (R/R(1-e)) \times (R/Re) \cong Re \times R(1-e)$$

Further given $re \in Re$, $ree = ere = re^2 = re$, so e is identity in Re , similarly $1-e$ is identity of $R(1-e)$. $\square$

Thm. Let R be a Ring with identity $1 \neq 0$.

Let $e \in R$ be an idempotent in R such that $er = re$ for all $r \in R$.

Then, Re and $R(1-e)$ are (two-sided) ideal of R and that

$$R \cong Re \times R(1-e)$$

Moreover, e and $1-e$ are identity for Re and $R(1-e)$, respectively.

Proof. Notice e commutes with respect to all elements in R , $Re = eR$.

and given $r_1, r_2 \in R$, $r_1e - r_2e = (r_1 - r_2)e \in Re$, so Re is subgroup

and given $r \in R$ and $r'e \in Re$, $rr'e \in Re$, so Re is left-sided ideal.

Since $eR = Re$, similarly it is right-sided ideal.

And, since $(1-e)^2 = 1-e+e-e^2 = 1-e$, so $1-e$ is also idempotent and

$1-e$ also commutes with all elements in R , so $R(1-e)$ is also ideal.

And, given $re \in Re$, $re \cdot e = re = e \cdot re$, so e is identity of Re ,

and given $r(1-e) \in R(1-e)$, $r(1-e) \cdot (1-e) = r(1-e+e-e^2) = r(1-e) = (1-e)r(1-e)$,

so $1-e$ is an identity of $R(1-e)$.

Define a map $\mathcal{Q}: R \rightarrow Re \times R(1-e) : r \mapsto (re, r-re)$. Then,

$$\begin{aligned}\mathcal{Q}(r_1 + r_2) &= (r_1 + r_2)e, (r_1 + r_2) - (r_1 + r_2)e \\ &= (r_1e, r_1 - r_1e) + (r_2e, r_2 - r_2e) = \mathcal{Q}(r_1) + \mathcal{Q}(r_2) \\ \mathcal{Q}(r_1 r_2) &= (r_1 r_2 e, r_1 r_2 - r_1 r_2 e) \\ &= (r_1 e r_2 e, (r_1 - r_1 e)(r_2 - r_2 e)) = \mathcal{Q}(r_1) \mathcal{Q}(r_2)\end{aligned}$$

Therefore, $\mathcal{Q}$ is Ring-Homomorphism and onto, because:

given $(r_1e, r_2(1-e)) \in Re \times R(1-e)$, let $r = r_1e + r_2(1-e)$. Then,
 $\mathcal{Q}(r) = (r_1e^2 + r_2(1-e) \cdot e, r_1e(1-e) + r_2(1-e)^2) = (r_1e, r_2(1-e))$.

Moreover, $\ker \mathcal{Q} = \{r \in R \mid \mathcal{Q}(r) = (0, 0)\}$, and $re = 0$ and $r(1-e) = 0$ implies
 $re + r(1-e) = r = 0$, so $\ker \mathcal{Q} = \{0\}$. So the first isomorphism theorem
gives that $R / \ker \mathcal{Q} \cong R \cong Re \times R(1-e)$. $\square$

Prop. If R and S are non-zero rings, then $R \times S$ is never a field.

Proof. Define a map

$$\mathcal{Q}: R \times S \rightarrow R: (r, s) \mapsto r.$$

Then, $\mathcal{Q}((r_1, s_1) \cdot (r_2, s_2)) = \mathcal{Q}(r_1 r_2, s_1 s_2) = r_1 r_2 = \mathcal{Q}(r_1, s_1) \mathcal{Q}(r_2, s_2)$,
and $\mathcal{Q}((r_1, s_1) + (r_2, s_2)) = \mathcal{Q}(r_1 + r_2, s_1 + s_2) = r_1 + r_2 = \mathcal{Q}(r_1, s_1) + \mathcal{Q}(r_2, s_2)$.

So, $\mathcal{Q}$ is Ring Homomorphism and trivially onto.

Moreover, $\ker \mathcal{Q} = \{(r, s) \in R \times S \mid r = 0\} = \{0\} \times S$.

So, $R \times S$ has non-trivial proper ideal, so it is never a field. $\square$